

SYAMA | SAG PEBBLE (SCAT) ORE SORTING

Assessment of the business case and recommendation on pebble ore sorting

Field	Detail	Field	Detail
Date	5 August 2026	Basis	African gold mine study (2024), re-parameterised
Subject	Sorting of SAG scats, converted (ex-oxide) line	Key input	Scat grade 2.0 g/t Au (client sampling)

1. Recommendation

Conditional GO — fund test work, not the plant. On the reference study's basis the case returns an NPV of US\$55.2 M with a 5.3-month payback on US\$4.1 M of capital, but the entire result rests on one assumption that has not been tested: that freed SAG capacity gets refilled with ore. Strip that away and the same plant destroys US\$36.9 M. Breakeven needs 40% of the freed capacity used. Two gates must close before any capital is committed.

- Gate 1 (commercial, no cost, ~4 weeks) — confirm that comminution, not the roaster/flotation circuit or ore supply, is the binding constraint on the converted line post-SSCP, and that ore exists to fill it.
- Gate 2 (technical) — STARK Phase 2 mill pebble test campaign on 2-5 t of Syama scats, sensor selection open. A 2.23× mass contrast is not proof of a detectable sensor contrast.

Do not order a sorter until both gates pass. The downside case is not marginal — it is a materially negative NPV plus permanent gold loss to the waste dump.

2. What the work to date establishes

The 2024 African gold mine study simulates a pebble sorter on the SAG mill discharge screen oversize. Barren scat mass is rejected, the circulating load falls, freed SAG capacity is refilled with ROM, and mill/leach feed tonnage is held constant by Goal Seek so the benefit reports as higher feed grade. At that operation it produced +7,925 oz/a for US\$8.2 M. The engine is sound and transfers directly to Syama, and its ratios and unit rates — dilution, scat load, sorter duty, costs, discount rate — are carried across in place of site data we do not yet have. What does not transfer is the material property that made the reference case work.

Parameter	Reference	Syama	Why it matters
ROM / scat grade (g/t Au)	1.57 / 0.55	2.40 / 2.00	Reference scats were barren; Syama's are not
Scat grade as % of ROM	35%	83%	Gold is not rejecting to the coarse fraction
Heterogeneity contrast	8.91×	2.23×	Back-solved waste:ore deportment to scats — the crux
Barren mass available in scats	~69%	~27%	Hard ceiling on what any sorter can reject
Reject grade at 90/90 duty (g/t)	0.151	0.782	Gold walking out to the dump
Mass pull achieved	54%	27%	Half the mass rejection, at five times the grade

3. Assessment — what a 2 g/t scat grade means

- The scats are not a waste stream. At 2.0 g/t they run at 83% of ROM grade and carry 14% of the gold in the circuit. Back-solving the deportment on the reference study's dilution gives a waste:ore contrast to scats of only 2.23× — against 8.91× in the reference case. Barely a quarter of the scat stream is barren dilution; that is the hard ceiling on mass rejection.
- Every rejected tonne is expensive. At 0.78 g/t and US\$4,000/oz the reject stream holds US\$75/t of recoverable gold against the US\$7.69/t of mining, dumping and G&A it avoids — it discards nearly ten dollars of gold for every dollar saved. The reference case discarded 0.151 g/t on the same duty.
- The comminution problem is already being solved. The SSCP installs a secondary crusher and a pebble crusher specifically to handle scats. Crushing recovers 100% of the contained gold; sorting discards some of it. Sorting must beat crushing, not merely beat doing nothing.
- Nothing here is a recovery gain. Scats recirculate — no gold is currently being lost. The value is purely throughput debottlenecking, which is worth exactly nothing if the SAG is not the binding constraint.

4. Economics — and the single factor that decides them

Scenario (reference sorter duty: 90% ore rec / 90% waste rej)	Extra oz/a	Net US\$/a	NPV @8%
Freed capacity fully refilled with 2.4 g/t ore	+2,485	+9.34 M	+55.2 M
Half refilled	+667	+2.09 M	+9.1 M
NPV breakeven	+303	+0.63 M	0 (40% capture)

Scenario (reference sorter duty: 90% ore rec / 90% waste rej)	Extra oz/a	Net US\$/a	NPV @8%
No spare ore — ROM held flat	−1,151	−5.17 M	−36.9 M
Refilled with 1.2 g/t stockpile instead	+667	+2.07 M	+9.1 M

Adopting the reference basis widens the spread rather than narrowing it: the upside rises to US\$55 M, but so does the cost of being wrong, and breakeven moves out to 40% capture. Secondary sensitivities stay mild — at US\$2,500/oz the case still returns +US\$31.5 M, and no sorter setting changes the verdict. The scat grade remains the dominant material lever: at 0.8 g/t the same plant would be worth US\$117 M. At 2.0 g/t, more than half the reference-style prize has already been lost to the ore's own character.

If the scat grade were...	Contrast	Reject g/t	Mass pull	NPV @8%
0.8 g/t (reference-like)	9.5×	0.22	55%	+117.3 M
1.2 g/t	5.7×	0.35	44%	+100.7 M
1.6 g/t	3.7×	0.52	36%	+80.4 M
2.0 g/t — Syama, as measured	2.2×	0.78	27%	+55.2 M
2.4 g/t (= ROM grade)	1.0×	1.31	19%	+22.9 M

Read down that table: the scat grade alone has removed more than half the prize before a single sorter is specified, and it does so by making the reject stream progressively more valuable. Sorter duty is the weaker lever — backing off to 99/50 still returns US\$34 M, and pushing to 85/95 gains under US\$1 M because the gold lost outruns the extra mass rejected. There is no sorter setting that rescues a poor contrast.

Against STARK's published gold case — 570 t/h at 1.5 g/t, NPV US\$79 M on US\$8 M — Syama matches the NPV per tonne only because it is run at US\$4,000/oz; on the benchmark's own price basis it sits well below. It frees 4.3% of mill feed against the “up to 15%” headline: a quarter of the prize.

5. Why Gate 1 is the real risk

Public disclosure points away from comminution being the constraint. Sulphide stockpiles were drawn down through 2025 to maintain throughput — a mill short of ore, not short of capacity. The SSCP is adding 60% of sulphide capacity (2.4 → 4.0 Mtpa), and the declared debottlenecking target is the roaster and its new ESP, not the mills. If flotation, the roaster or ore supply binds first, every tonne of freed SAG capacity is worth zero and the case collapses to the −US\$36.9 M row above. This must be settled from the plant's own constraint data before anything else is spent.

6. Risks and adjacent opportunity

- Refractory gold — reject grades must be confirmed by fire assay and diagnostic leach, not sensor response. XRT sees density, not gold, and sensor selection across colour, NIR, XRT, XRF and EM is itself an open question.
- Sulphidic rejects — geochemical/ARD characterisation is required before a new reject stream is sent to a waste dump; this can carry its own closure cost.
- Mali country risk — supply chain and spares disrupted through 2026, and single-sorter redundancy is nil. The reference study's 8% discount rate is used here; a Mali-specific premium would cut the upside case by roughly a tenth.
- Larger prize elsewhere: SLC dilution rejection on ROM or coarse-crushed underground feed. Some 20% dilution across ~2.6 Mt/a offers far more heterogeneity than the scats do, and the same sample campaign can test it at marginal extra cost. Recommend it be scoped into Gate 2.

7. Next steps

#	Action (STARK phased approach)	Owner	Cost	Duration
1	Constraint audit + Phase 1 desktop study: is comminution binding post-SSCP, is there ore to fill it, and close out mass balance, PSDs, bond indices	Site metallurgy / STARK	<\$20 k	4 weeks
2	Phase 2 mill pebble test campaign — 2-5 t, hardness and sortability by rock type	STARK (OEM-agnostic)	\$60-150 k	3 months
3	Phase 3 online analyser — scan-only sorter in circuit, no ejection, no gold at risk	STARK / site	tbc	3-6 months
—	ARD characterisation of the reject stream	Environmental	<\$30 k	2 months
4	Re-run this model on measured response; decide on Phase 4 capital	Study team	nil	2 weeks

Model: Syama_Pebble_Sorting_Simulation.xlsx. All costs in US\$. Where Syama site data was unavailable, ratios and unit rates are carried from the 2024 African gold mine study: dilution 11.9%, scat load 16.4% of new feed, sorter duty 90/90, US\$7.69/t mining-dump-G&A on rejects, US\$0.45/t sorting, 7,370 h/a, 8% discount, 10-year life. Syama's own ROM grade, recovery, gold price and measured scat grade are retained. The 11.9% dilution is the one carry-over we would question — STARK's own gold benchmark uses 15%, and dilution drives the contrast (Sensitivity table E).